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B.Tech. Degree IV Semester Examination April 2019

ME 1406 MANUFACTURING PROCESS (2012 Scheme)

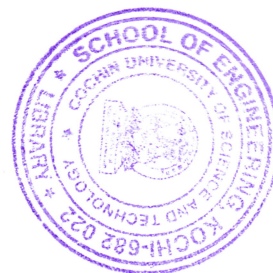
Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) What are the types of cores used in casting process?
- (b) List the various pattern allowances and write a short note on them.
- (c) What do you mean by mould designing for economic moulding?
- (d) What are the features which aid handling in a casting operation?
- (e) Explain swaging operation.
- (f) Write a short note on rolling stand arrangements.
- (g) Explain TIG welding.
- (h) What is electro slag welding?



PART B

(4 × 15 = 60)

- II. (a) Explain modulus method for finding the optimum riser size. (10)
- (b) What are the forces acting on moulding flasks? (5)
- OR**
- III. (a) Describe the various types of patterns used in casting process. (10)
- (b) What are the advantages and limitations of metal casting process? (5)
- IV. (a) Write a short note on casting metallurgy. (5)
- (b) Explain precision investment casting with a neat sketch. What are its advantages and disadvantages? (10)
- OR**
- V. (a) Explain centrifugal casting with neat sketches. (10)
- (b) Write a short note on defects in casting. (5)
- VI. Explain smith, drop and press forging with neat sketches. (15)
- OR**
- VII. Explain shearing, bending and stretch forming operation with neat-sketches. (15)
- VIII. (a) Write a short note on laser beam welding and thermit welding. (10)
- (b) Differentiate between brazing and soldering. (5)
- OR**
- IX. (a) With the help of a neat sketch explain arc welding equipment. (10)
- (b) Explain projection welding. (5)
